

1. (a) Implement a single-layer perceptron for binary classification using PyTorch.
(b) Write a program to classify data from three different classes using a multi-layer neural network.
2. (a) What are the key differences between a single-layer and multi-layer feedforward neural network?
(b) Implement backpropagation manually (without using PyTorch autograd).
3. (a) Using Amazon Food Review data, classify reviews as positive or negative.
(b) Write a PyTorch program to classify IMDb or Amazon reviews using an LSTM.
4. (a) Train an ANN to classify the MNIST dataset. Report training accuracy and test accuracy.
(b) Explain the vanishing gradient problem in RNNs and demonstrate using simple RNN and LSTM.
5. (a) Demonstrate how to install Python, PyTorch, and upload files to Google Colab.
(b) Create a pipeline to download and preprocess a Kaggle dataset for training a binary classifier.
6. (a) Use CNN to classify CIFAR-10 images and plot training/validation accuracy.
(b) Perform transfer learning for emotion detection using a pre-trained CNN.
7. (a) Modify your CNN to include batch normalization and dropout.
(b) Modify your CNN architecture to include residual connections (similar to ResNet).
8. (a) Apply transfer learning using ResNet18 or VGG16 for image classification.
(b) Compare the performance of a custom CNN vs a pre-trained CNN on the CIFAR-10 dataset.
9. (a) Visualize attention in RNNs using sentiment analysis.
(b) Describe and compare CNN-based and RNN-based architectures for image captioning.

- 10.** (a) Implement a simple GAN to generate handwritten digits (MNIST).
(b) Create a GAN to generate synthetic images and track training progress with loss curves.#
- 11.** (a) Demonstrate use of basic data augmentation for CIFAR-10 dataset.
(b) Implement data augmentation techniques using torchvision.transforms and compare performances.
- 12.** (a) Write code to save and load PyTorch model weights.
(b) Explain and demonstrate early stopping during training.
- 13.** (a) Compare binary and multi-class classification problems.
(b) Implement a confusion matrix and classification report for any dataset.
- 14.** (a) Fine-tune a pre-trained MobileNet or EfficientNet on a small custom dataset.
(b) Build a text preprocessing pipeline using torchtext or nltk for Amazon Food Reviews.
- 15.**(a) Train a CNN model on CIFAR-10 and visualize feature maps from convolutional layers.
(b) Implement a Bidirectional LSTM for text classification and compare with unidirectional LSTM#